

Project Report - Challenge: Urban Mobility

kemptAInability - An interactive traffic flow simulation focusing on
the city of Kempten

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Contents

1	Introduction	1
2	Urban Mobility Transformation in Kempten	1
2.1	Context and Problem Definition	1
2.2	Policy and Regulations	2
2.3	Challenge	2
3	Concept	2
3.1	Scenarios	3
3.2	Color and Functionality Concept	3
3.3	Implementation	3
3.3.1	Technology and chosen software	3
3.3.2	Features and Insights.	4
4	Business Perspective	4
5	Conclusion and Outlook	4

1 Introduction

Urban areas are facing increasing pressure due to rising traffic volumes, noise levels, and greenhouse gas emissions, increased by a limited amount of green spaces to balance the urban environment. According to the European Parliament, Urban mobility is responsible for around 70% of total air pollutants and in particular, for around 40% of CO₂ emissions in cities, although the resident's physical and mental health can be weakened by such air and noise pollution [10].

In the city of Kempten (Allgäu) a major cause of high traffic volume is the B19 highway that runs right through the city center which is an important route to commuters and residents. However, Kempten expects an upcoming infrastructural shift due to reconstructions. The St.-Mang-Brücke, a central bridge that carries traffic of the B19 highway, is planned to be closed for two years [6]. This change leads to concerns among frequently drivers as well as opportunities in urban mobility planning. Residents are afraid of increased traffic delays and stress during their rides, while the city's mobility management sees the potential to reroute traffic more sustainably and reduce inner-city pollution and noise [17].

In response, our team developed an interactive traffic simulation that will be presented in the city's Future Lab [16]. The aim of this tool is to inform citizens, and encourage their sustainable choice of transportation while experimenting with real-world traffic scenarios in a visual and intuitive way. Furthermore, the application is to convince political stakeholders to recognize the impact of traffic in the city and integrate greener alternatives into their proactive decisions.

By integrating low- and peak-hour traffic data, the simulation provides users with a clear visualization, not only containing traffic delays, but also health benefits, such as reductions in CO₂ emission and traffic noise, when specific streets are closed by barriers and cars get redirected.

2 Urban Mobility Transformation in Kempten

2.1 Context and Problem Definition

Kempten is a city where tradition shapes many aspects in the life of residents, including the way of movement through the urban areas. For a large part of the population, the private car remains the standard and preferred mode of transport which is deeply integrated into daily routines. The environmental consequences of too much air and noise pollution and too less green areas in the inner-city are often overlooked. This is not mainly due to indifferences, but rather because alternatives have rarely been offered in

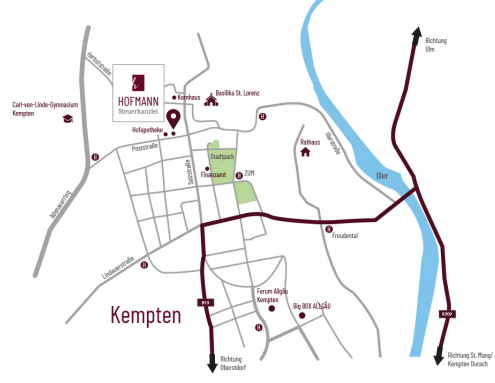


Figure 1: City Center of Kempten

recent years. Recognizing this challenge, the mobility management department of Kempten has invested in an improved public transport infrastructure (ÖPNV) [5]. However, Stefan Sommerfeld, who is part of the mobility management of Kempten mentioned that these developments alone will not be sufficient to bring a significant shift in public behavior [17]. Therefore, the remaining question is:

How can residents be motivated to consider the impacts of their mobility choices on the environment, physical health, and the quality of everyday life?

This behavior is becoming an unintentional change by the reconstruction of St.-Mang-Brücke. The central bridge that is currently part of the B19 federal highway directly through the city center (see Figure 1), is scheduled for full closure during 2 years [7]. The planned closure of a main part of B19 highway requires a rethinking of how traffic can circulate around the city core. Furthermore, inner-city residents experience a reduction of air and noise pollution and traffic which leads to a cleaner, quieter and more livable city. On the other side, the construction work poses a logistical difficulty for many residents and commuters. They are concerned about more traffic delays and longer travel routes [17].

A digital tool is the most pleasing way to inform the residents of the impact as it makes complex issues relatable. Traditional approaches, such as SUMO (Simulation of MObility) [11] or Vissim and Visum (Simulations by PTV GROUP) [15], developed traffic simulations that allow modeling for route finding and reconstruction of transportation systems. Therefore, they can measure traffic volumes and emissions under different circumstances [11, 15]. Unfortunately, these applications are too technical for usual users and require expert knowledge to interact with. A promising innovation for traffic simulation is the digital twin technology. It is a personalized, digital copy of the city of Kempten to optimize traffic efficiency and support sustainable urban mobility. They can visualize predictive analytics and simulate the performance of the



Figure 2: Planned Reconstruction Design of St.-Mang-Brücke

transportation network [12]. Nevertheless, its cost- and time-expensive development, make it not yet an option for the city.

2.2 Policy and Regulations

As mentioned before, no rerouting of the B19 federal highway is currently planned. Instead, the city intends to reconstruct the St.-Mang-Brücke with the same amount of car lanes for maintaining the central route through the city. The planned design that can be seen on Figure 2, continues to prioritize inner-city traffic over structural transformation. The reason for this is that B19 highway belongs to the federal state of Bavaria and not the city of Kempten.

However, the urban mobility and infrastructure planning of mobility management in Kempten is still planning a sustainable transformation. The idea is influenced by given policy frameworks such as the Sustainable Development Goal (SDG) from the United Nations. Specifically, matching with SDG 11: Sustainable Cities and Communities, which emphasizes the need for safe, resilient, inclusive, and sustainable urban planning [18]. On addition, the European Green Deal states a substantial reduction in CO2 emissions, where traffic is a big trouble maker. They also promote a climate-resilient infrastructure and resident involvement in urban transformation processes [9]. These policies and regulations not only shows the importance of resident information and integration in the planning process, but also the need to transform Kempten towards a more sustainable and greener urban city.

2.3 Challenge

The challenge lies in transforming the planned closure of the St.-Mang-Brücke from a logistical transformation into an opportunity for sustainable behavioral change through a resident information. This requires developing an accessible, data-driven tool that empowers residents to understand and engage with the environmental and health impacts of their transportation choices, while influencing policy-makers to support greener alternatives.

In summary, the following core objectives must guide the development:

- Usage of real-world data in low- and peak-hours to simulate different traffic times.
- Accessible to non-experts and usable for everyone with focus on user-centered design to be monitored in Kemptens Future Lab.
- More affordable than a digital twin but limited to the most relevant facts such as the impact of less traffic in the inner-city.
- Visualization of structural impacts, such as traffic delays, and environmental impacts, such as CO2 emissions and noise heatmaps.
- Serious Gamification features such as interactions with road closures.
- Inform residents for upcoming infrastructural shifts such as bridge closure.
- Foster residents in active decision-making towards more conscious, sustainable transportation choices.
- Convince political stakeholders to contribute proactive to Kempten’s life quality.

3 Concept

While developing our concept, we initiated primary research and talked to our contact persons, among others, to find out specific requirements and design an optimal project for the city, factoring in the diverse needs of all stakeholders alongside our insights into effective and enjoyable development practices. Following the idea generation phase, as elaborated in preceding sections, our project is centered around the creation of an innovative application aimed at two primary target groups—political stakeholders and residents of Kempten. This application will demonstrate how alternative transport routes can enhance the city’s quality of life while enhancing a more sustainable environment. To emphasize traffic scenarios from the residents’ perspective and identify their most significant pain points, we developed two personas: Hanna and Claire. Hanna is concerned about the implications of the bridge closure and is the niece of Claire, who serves on the city council. Together, we analyzed potential scenarios that might encourage Hanna to collaborate with the city council, while also considering the inquiries a Kempten resident, like Claire, might pose to a city council member like Hanna. This process, along with discussions with our contact person, facilitated the simulation of two critical scenarios that could significantly influence political opinion leaders.

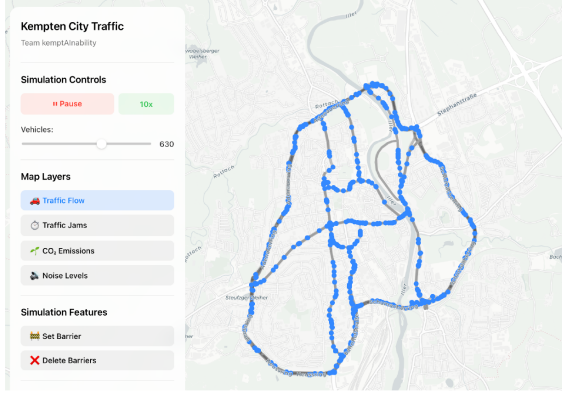


Figure 3: Traffic Flow

3.1 Scenarios

One scenario focused on the closure of the St.-Mang-Bridge, a decision that initially encountered resistance from various interest groups. However, we quickly recognized the viability of the ring road as an alternative. Our visualizations illustrated a reduction in both noise pollution and emissions, highlighting the feasibility of this option. Furthermore, redirecting traffic from the city center to the ring road may result in an increase in travel time of only 1 to 2 minutes. This minor delay is a negligible trade-off for the substantial benefits of a greener urban environment characterized by significant reductions in noise and emissions. Additionally, we propose conducting individual interactive experiments with our system, such as a second scenario—the closure of Residenzplatz—to evaluate the subsequent alterations in traffic dynamics and environmental parameters. This interactive approach to traffic investigation employs storytelling techniques to effectively convey planning insights, highlighting their potential positive impact on the community.

3.2 Color and Functionality Concept

Next, we present an overview of the color and interactive functionality concept that was subsequently implemented, utilizing sample images from the front end. Different traffic states are demonstrated using certain colors.

In the menu on the left-hand, as illustrated in Figure 3, users can initiate the simulation, select the number of vehicles using a slider, and simulate a specific traffic state within the map layers. Figure 3 displays the normal traffic flow in blue, with 630 cars traveling at 10 times the average speed, adjustable manually.

Moreover, colors such as orange and dark red are employed to represent traffic jams (see Figure 4, left), indicating areas with a high concentration of vehicles, which can be activated via the traffic jam button in the menu. The intensity of red indicates the intensity of

traffic jam: the darker the red, the more intensity. By selecting the CO2 emissions button, users can visualize areas with particularly high emissions (orange) versus those with lower emissions (green) see figure 4, center. The blue areas on the right in (figure 4, right) illustrate noise intensity, correlated to their size.



Figure 4: Left -traffic jam , center - emissions, right - noise impact

The 'Set Barrier' button allows for the placement of barriers on both sides of the roadway, enabling observation of their effects on traffic flow, noise, emissions, and traffic jams across different streets. Hovering over the road segments provides detailed road information, including name, length, width, speed limit, priority, authorized vehicles, and current status regarding closures.

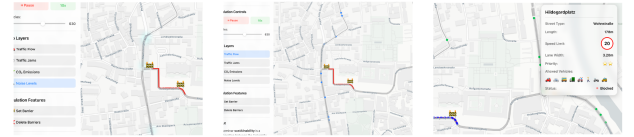


Figure 5: Demonstration of set barrier functionality

Through our project, we aim to convince both political stakeholders and citizens of Kempten to embrace a greener urban vision, featuring increased public green spaces, potential café installations, and expanded bike paths that would contribute to a more vibrant urban atmosphere. By implementing our project, we want to showcase the transformative potential of data in urban planning and its impact on community well-being.

3.3 Implementation

3.3.1 Technology and chosen software

To extract and process urban street network data for Kempten, we utilized Overpass Turbo [2] with a custom query specifically tailored to filter OpenStreetMap (OSM) data [14] based on a targeted set of street names. This query generated XML output containing relevant ways tagged as *highway*, along with their associated nodes and complete metadata, optimized for further processing.

The raw XML data underwent an initial preprocessing stage using JOSM [1] for manual inspection and pruning. Subsequently, custom scripts were applied within a dedicated virtual environment to systematically remove unnecessary elements, including pedes-

trian paths, cycleways, junction nodes, and traffic signals, ensuring a clean and focused dataset.

This refined dataset was converted into a SUMO (Simulation of Urban MObility) [11] compatible format utilizing SUMO’s `netconvert` tool, enabling the creation of a valid `.net.xml` file. Additional network simplification and manual refinements were performed using SUMO’s built-in editor, `netedit`, further optimizing the network structure.

In preparation for advanced traffic modeling integration, we experimented with TSMM (Transportation Systems Modeling and Management) [3], thereby establishing groundwork for future application of reinforcement learning (RL) and machine learning (ML) techniques.

For frontend visualization and simulation, the SUMO-generated network data was transformed into GeoJSON format [8], enriched with additional lane-level attributes through customized scripts adapted from SUMO’s `net2geojson`. The data was then visualized using a React [13] and Vite [4]-powered frontend application, featuring dynamic simulation of traffic flow, the application of traffic barriers, and visual representations of congestion, noise pollution, and CO2 emissions through interactive heatmaps and color-coded map layers. The frontend also provides real-time vehicle counts, speed limits and more relevant meta-data, and supports animated time controls.

While this prototype was developed as a proof of concept within a limited timeframe, the overall methodology is structured to be easily adaptable for diverse urban planning scenarios and scalable to additional cities beyond Kempten.

3.3.2 Features and Insights.

Throughout the project, we implemented a modular, reproducible pipeline leveraging widely adopted open-source tools and libraries such as `osmnx`, `geopandas`, and `pyproj`, all documented within a public GitHub repository containing a detailed README for setup and extensibility. The frontend supports features like interactive simulation control, heatmap overlays, traffic manipulation (e.g., barrier placement), and real-time monitoring. Tangible interaction mechanisms allow users to apply touch-based input to simulate real policy changes directly on the map. This includes scenario customization tools enabling participants to close roads, test pre- and post-2029 configurations, and instantly observe impacts on traffic flow and environmental indicators.

To enhance user engagement and educational value, we introduced a layer of gamification: participants gain points for creating sustainable, efficient traffic configurations and are rewarded both visually and statistically for improvements in life quality (e.g., reductions in congestion, noise, and CO2 emissions). Unlike comprehen-

sive digital twins, our interface deliberately simplifies data representation by focusing only on key indicators, avoiding overwhelming complexity while still supporting meaningful analysis, while being way more cost-effective than developing a digital twin from scratch.

During development, it became clear that OSM data—when enriched with real-time or domain-specific traffic data—offers significant potential for scalable urban mobility applications. The ability to dynamically select and process relevant OSM subsets enables tailored simulations for specific cities, districts, or even intersections, making this approach suitable for both generalized urban planning and specialized traffic engineering use cases. This insight underscores OSM’s value as a foundational data source for future smart mobility systems augmented by AI-driven optimization strategies.

4 Business Perspective

Unique Selling Points (vs Digital Twin)

Cheaper: Reduced modeling complexity Accessible: No technical background needed Intuitive: Easy to use for citizens and politicians alike Focused: Shows only relevant data — reduces cognitive overload Engaging: Narratives and gamification drive participation Target Stakeholders

Local governments (e.g. Kempten) Regional traffic planners Civic engagement labs Citizens & commuters Vision

Expand to other cities facing major rerouting or infrastructure updates Modular format allows adaptation to different city data sets Potential integration with: Online versions School education modules Urban hackathons

$$EyeAspectRatio = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2 \|p_1 - p_4\|} \quad (1)$$

5 Conclusion and Outlook

KemptAInability demonstrates how abstract infrastructure changes become graspable with tangible, AI-driven tools. Data democratization = better policy acceptance and smarter cities. As 2029 approaches, we plan to: Integrate broader datasets (e.g. seasonal variations) Launch online versions of the simulation Host participatory planning events Final message: “The St.-Mang-Brücke closure isn’t a crisis — it’s a catalyst.”

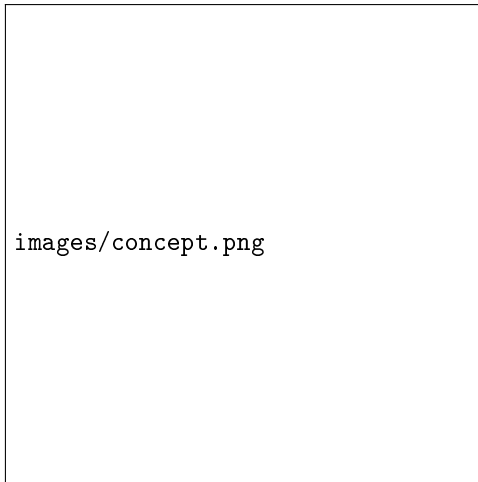


Figure 6: Grundlegende Schritte der Implementierung von *EmotionRate*

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